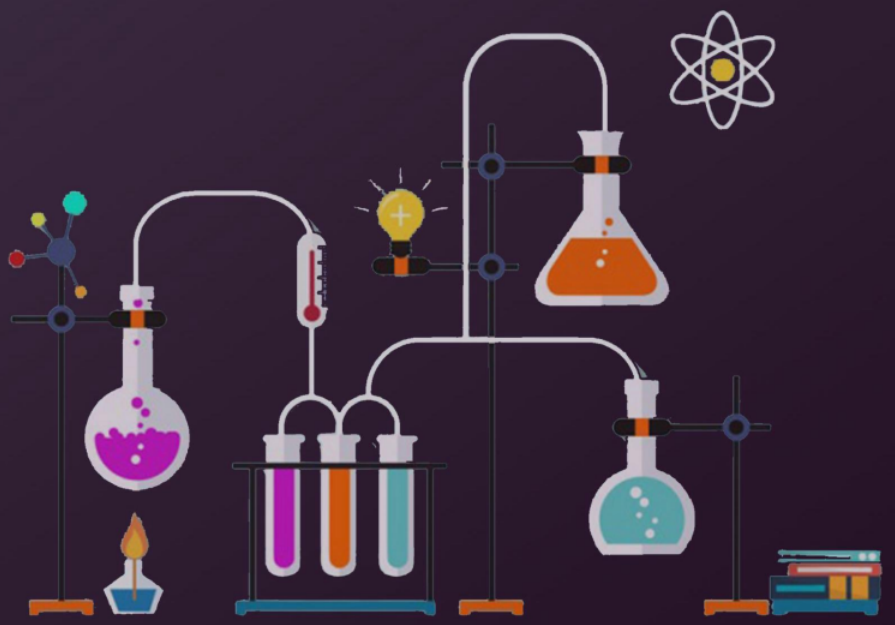
CHEMISTRY

# Carboxylic Acid



SHEET

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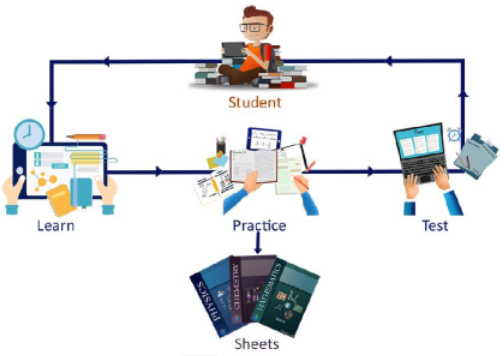
# *Contents*


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***Note***

*Detailed solutions are available on the eSaral App.*

# *Module Description*




For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

*"Practice makes a man perfect"*

**1. Concept builder – 1 & 1A**

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

**2. Brain Booster – 2 & 2A**

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

**3. Simulator – JM & JA**

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

# *Home Work Index*

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

| Sr. No. | Topics | Ex-1 | Ex-1A | Ex-2 | Ex-2A |
|---|---|---|---|---|---|
| 1 | Preparation and Physical Properties of Carboxylic Acids | 1-4 | 1 | 1-8 | 1, 13 |
| 2 | Chemical Properties of Carboxylic Acid | 5-9 | 2-4 | 9-10 | |
| 3 | Heating Effect of Hydroxy, Halo and Dicarboxylic Acids | 10-12 | 14, 17-18 | | 11, 14-15 |
| 4 | Wolff Rearrangement, Formic Acid and Acetic Acid | | | | |
| 5 | Preparation Methods for Amine | 13-19 | | 11-14 | |
| 6 | Chemical Properties of Amine, Carbylamine and Mustard Oil Test | 20-23 | 5, 9, 19-22 | 15-21 | 2-3, 16 |
| 7 | Diazotisation, ESR and Test of Aniline | 24-26 | 6, 10 | 22-30 | |
| 8 | ACC | 27-29 | 7-8, 11-13, 15-16, 23 | 31-36 | 4-10, 12 |

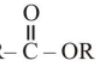
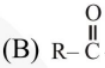
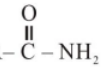
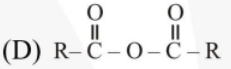
## Exercise - 1 — Concept Builder

### SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

### Preparation and Physical Properties of Carboxylic Acid

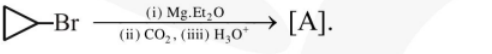
**Q1.** Hydrolysis of which of the following gives acid at fastest rate:

(A) 
(B) 
(C) 
(D) 

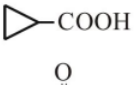
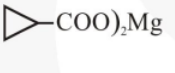
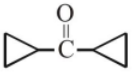
**Q2.** n-Butyl benzene on oxidation gives :-

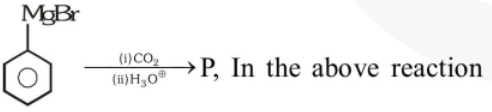
(A) Benzoic acid (B) Butanoic acid
(C) Benzyl alcohol (C) Benzaldehyde

**Q3.** For the following reaction,

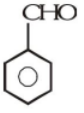
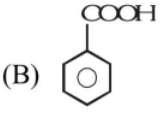
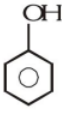
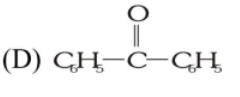


Product [A] is :

(A) 
(B) 
(C) 
(D) 

**Q4.** 

product 'P' is -

(A) 
(B) 
(C) 
(D) 

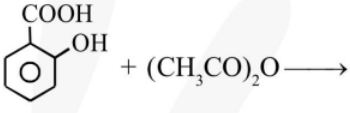
### Chemical Properties of Carboxylic Acid

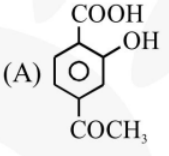
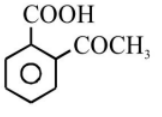
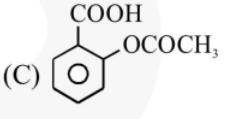
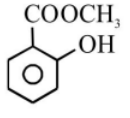
**Q5.** The rate of esterfication of HCOOH (I), $CH_3COOH$ (II), $(CH_3)_2CHCOOH$ (III) and $(CH_3)_3$ CCOOH (IV) with ethanol follows in the order :-

(A) IV > III > II > I
(B) I > II > III > IV
(C) II > I > IV > III
(D) III > IV > I > II

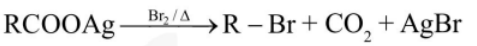
**Q6.** Acetic acid reacts with $PCl_5$ to give-

(A) $Cl_2CHCOOH$ (B) $CH_3COCl$
(C) $ClCH_2COOH$ (D) $ClCH_2COCl$

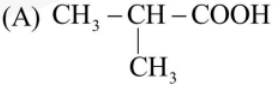
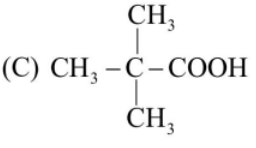
**Q7.** 

(A) 
(B) 
(C) 
(D) 

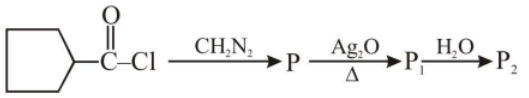
**Q8.** Consider the given reaction

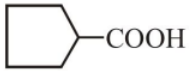


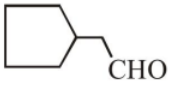
which one of the following acid will give maximum yield of R–Br in the above reaction?

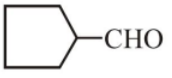
(A) 
(B) $CH_3–CH_2–CH_2–COOH$
(C) 
(D) All will give same yield

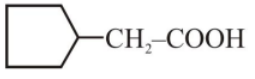
**Q9.** Identify $P_2$ in the given reaction sequence -



(A) 

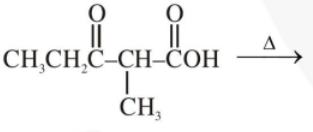
(B) 

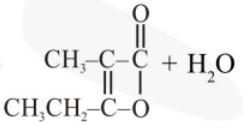
(C) 

(D) 

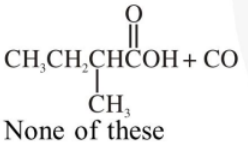
## Heating Effect of Hydroxy, Halo and Decarboxylic Acid

**Q10.** The products obtained in the reaction:

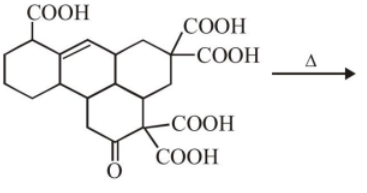


(A) 

(B) 

(C) 

(D) None of these

**Q11.** 

Number of $CO_2$ evolved.

(A) 5

(B) 2

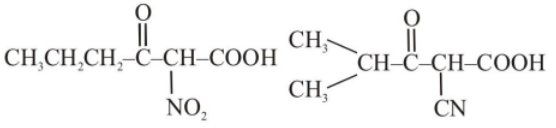
(C) 3

(D) 4

**Q12.** Correct order of decarboxylation



(a) (b)



(c) (d)

(A) a > b > c > d (B) c > d > b > a

(C) c > d > a > b (D) d > c > a > b

## Preparation Methods for Amine

**Q13.** A reaction used in descending a homologous series would be -

(A) $RCONH_2 + Br_2 + KOH$

(B) $RCH_2Cl + KCN$

(C) $RNH_2 + CHCl_3 + KOH$

(D) None of the above

**Q14.** Gabriel pthalimide reaction is used in the preparation of :-

(A) Secondary amine

(B) Primary aliphatic amine

(C) Primary aromatic amine

(D) Teritiary amine

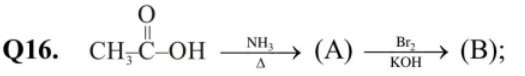
**Q15.** Ethanamine can be obtained if the following compound is heated with [KOH + $Br_2$] :

(A) Ethanamide

(B) Methanamide

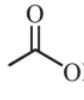
(C) Propionamide

(D) All of the above

**Q16.** 

Product (B) is -

(A) $CH_3–ND_2$ (B) $CH_3–NH_2$

(C)  (D) 

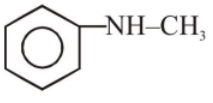
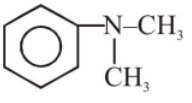
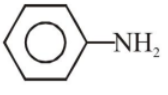
**Q17.** Hofmann degradation is useful for :-
(A) Stepping up any homologous series
(B) Stepping down any homologous series
(C) Both the above
(D) None of the above

**Q18.** N-Ethyl pthalimide on hydrolysis gives:
(A) Methyl alcohol (B) Ethyl amine
(C) Dimethyl amine (D) Diethyl amine

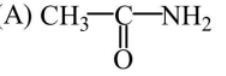
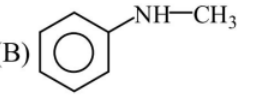
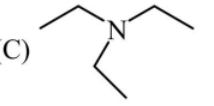
**Q19.** Consider the following sequence of reaction
$C_2H_5$–Br $\xrightarrow{AgCN}$ A $\xrightarrow{H_2/Ni}$ B Product B is
(A) $CH_3CH_2CH_2$–$NH_2$
(B) $CH_3$–$CH_2$–NH–$CH_3$
(C) $(CH_3CH_2)_2NH$
(D) $CH_3$–$CH_2$–$NH_2$

## Chemical Properties of Amine, Carbylamine and Mustard Oil Test

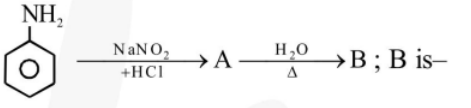
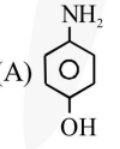
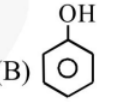
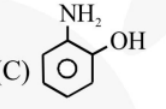
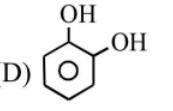
**Q20.** $C_6H_5NH_2$, $CHCl_3$ and KOH give the main product :-
(A) Phenyl cyanide
(B) Benzyl Cyanide
(C) Aniline
(D) Phenyl isonitrile

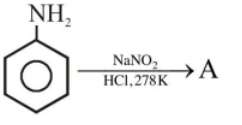
**Q21.** Carbylamine reaction is positively given by :
(A) 
(B) 
(C) 
(D) $Me_2NH$

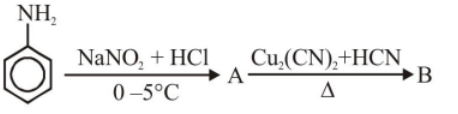
**Q22.** Ethyl amine on heating with $CS_2$ in presence of $HgCl_2$ forms :-
(A) $(C_2H_5)_2S$ (B) $C_2H_5NCS$
(C) $(C_2H_5)_2CS$ (D) $(C_2H_5)(CS)_2$

**Q23.** Which of the following compound give foul smell when reacted with $CHCl_3$ + NaOH
(A) 
(B) 
(C) 
(D) 

## Diazotisation, ESR and Test Aniline

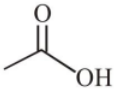
**Q24.** 
(A) 
(B) 
(C) 
(D) 

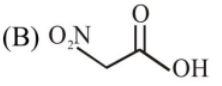
**Q25.** In the chemical reactions  B, the compounds A and B respectively are:
(A) Fluorobenzene and phenol
(B) Benzene diazonium chloride and benzonitrile
(C) Nitrobenzene and chlorobenzene
(D) Phenol and bromobenzene

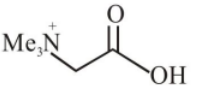
**Q26.** 
The compound A and B respectively are :-
(A) Nitrobenzene and chlorobenzene
(B) Nitrobenzene and phenol
(C) Chlorobenzene and Nitrobenzene
(D) Benzene diazonium chloride and benzonitrile

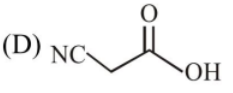
## ACC

**Q27.** Which of the following has the highest $pK_a$ value?

(A) 

(B) 

(C) 

(D) 

**Q28.** Which optically active compound on reduction with $LiAlH_4$ will give optically inactive compound?

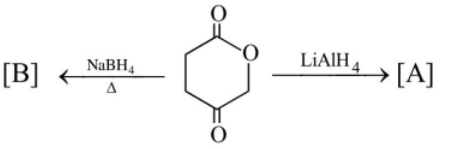
(A) $CH_3-\underset{\underset{OCH_3}{|}}{CH}-COOH$

(B) $CH_3-CH_2-\underset{\underset{OH}{|}}{CH}-COOH$

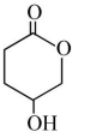
(C) $CH_3-CH_2-\underset{\underset{CH_2OH}{|}}{CH}-COOH$

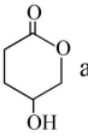
(D) $CH_3-\underset{\underset{OH}{|}}{CH}-CH_2-COOH$

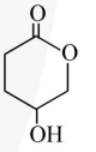
**Q29.** In the given reaction,



[A] and [B] respectively be :

(A) $CH_2OH-CH_2-CH_2-\underset{\underset{OH}{|}}{CH}-CH_2OH$ and 

(B)  and $CH_2OH-CH_2-CH_2-\underset{\underset{OH}{|}}{CH}-CH_2OH$

(C) Both are 

(D) Both are $CH_2OH-CH_2-CH_2-\underset{\underset{OH}{|}}{\overset{\overset{OH}{|}}{CH}}-CH_2OH$

## *Exercise - 1A* *Concept Builder*

### ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s).* ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

**Q1.** Which type of isomerism is/are exhibited by hexanoic acid.

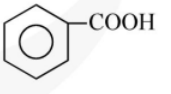
(A) Chain isomerism

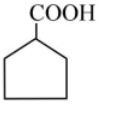
(B) Position isomerism

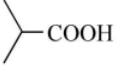
(C) Functional isomerism

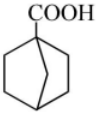
(D) Metamerism

**Q2.** Which one of the following compounds will give HVZ reaction?

(A) 

(B) 

(C) 

(D) 

**Q3.** RCOOR' can be prepared by :

(A) Esterification of RCOOH

(B) Reaction of $CH_3CH = CH_2$

(C) Baeyer-Villager oxidation of RCOR' with peroxy acid

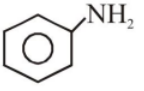
(D) Reaction of RCOCl with R'OH

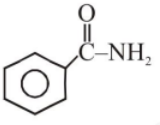
**Q4.** Which of the following will form acetyl chloride with $PCl_5$ ?

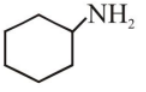
(A) MeCOOH (B) MeCOOMe

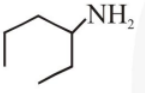
(C) MeCOOCOMe (D) Me–$CONH_2$

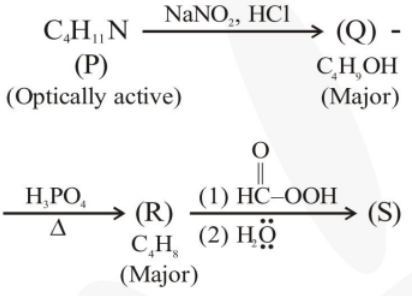
**Q5.** Which of the following compounds will produce a compound with offensive smell when treated with $CHCl_3$ / KOH ?

(A) 

(B) 

(C) 

(D) 

**Q6.** 

Correct statement regarding above sequence of reaction is/are :

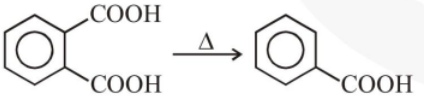
(A) Q give immediate turbidity with Anhydrous $ZnCl_2$ + HCl

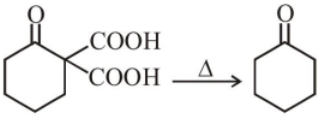
(B) R is cis But-2-ene

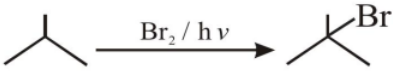
(C) Resolvable product will be form when R is treated with $OsO_4$ ; $H_3O^+$

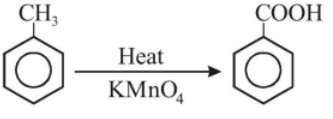
(D) S is a meso diol.

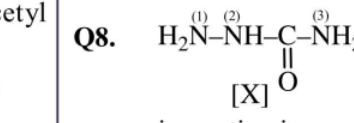
**Q7.** Select correct reaction among following :

(A) 

(B) 

(C) 

(D) 

**Q8.**  ; Which of the following option is correct regarding [X] ?

(A) [X] gives semicarbazone when reacts with propanal in basic medium

(B) $\overset{(1)}{N}$ is the donor atom

(C) [X] acts as base

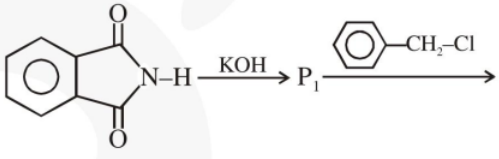
(D) [X] acts as nucleophile

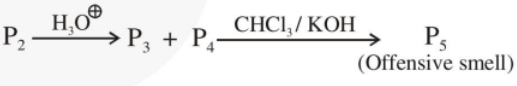
## PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

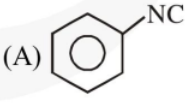
**Paragraph for Questions 09 & 10**

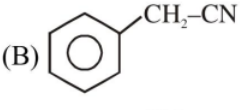
Gabrieal pthalimide is method of synthesis of primary amine by using following sequence.

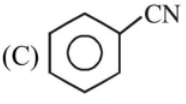


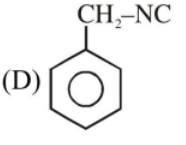


**Q9.** Product $P_5$ is-

(A) 

(B) 

(C) 

(D) 

**Q10.** Incorrect statement is-

(A) $P_4$ on reaction with $NaNO_2/HCl$ followed by β-napthol gives red colour dye.

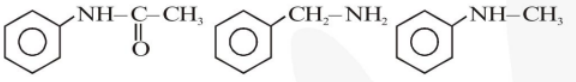
(B) $P_3$ on heating can form anhydride

(C) Conversion of $P_1$ to $P_2$ is example of $SN^2$ mechanism.

(D) Index of hydrogen deficiency (Double bond equivalent ) in $P_2$ is 11.

**Paragraph for Q.No. 11 to 12**

For given compounds.




(P) (Q) (R)

**Q11.** Basic strength order is -

(A) P > Q > R

(B) Q > R > P

(C) R > Q > P

(D) R > P > Q

**Q12.** Incorrect statements is-

(A) QR is a functional isomer pair

(B) PQ & PR are functional isomers respectively

(C) P, Q & R all are aromatic compound

(D) $Ph–\overset{\overset{O}{\|}}{C}–NHCH_3$ is a metamer of P

## MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

**Q13.**

**List-I**

(P) $CH_3–\overset{\overset{O}{\|}}{C}–OH \xrightarrow{PCl_5}$

(Q) $CH_3–\overset{\overset{O}{\|}}{C}–O–\overset{\overset{O}{\|}}{C}–CH_2CH_3 \xrightarrow{PCl_5}$

(R) $CH_3–\overset{\overset{O}{\|}}{C}–NH_2 \xrightarrow{PCl_5}$

(S) $CH_3–CH(\overset{\overset{O}{\|}}{C}–OCH_3) \xrightarrow{PCl_5}$

**List-II**

(1) $CH_3–CN$

(2) $CH_3–Cl$

(3) $CH_3–\overset{\overset{O}{\|}}{C}–Cl$

(4) $CH_3–CH_2–\overset{\overset{O}{\|}}{C}–Cl$

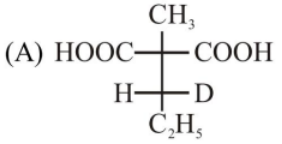
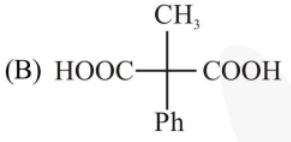
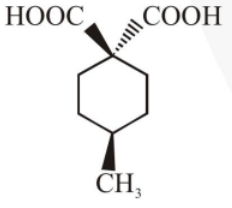
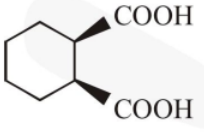
**Code :**

| | P | Q | R | S |
|---|---|---|---|---|
| (A) | 3 | 2,4 | 3,4 | 1 |
| (B) | 1 | 2 | 3 | 4 |
| (C) | 4 | 3,4 | 2,4 | 1 |
| (D) | 3 | 3,4 | 1 | 2,4 |

## MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

**Q14.** Match the following question :

| **Column - I (Reaction)** | | **Column-II (Products)** |
|---|---|---|
| (A)  | $\xrightarrow{\Delta}$ | (P) Diastereomers |
| (B)  | $\xrightarrow{\Delta}$ | (Q) Racemic mixture |
| (C)  | $\xrightarrow{\Delta}$ | (R) Meso comp. |
| (D)  | $\xrightarrow{\Delta}$ | (S) $CO_2$ gas will evolve |

**Q15.** Match the following question :

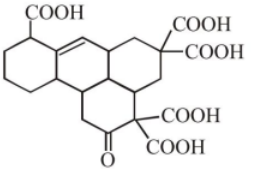
| **Column I (Organic compounds oxidised by $HIO_4$)** | **Column II (Products of $HIO_4$ oxidation)** |
|---|---|
| (A) $CH_3COCHO$ | (P) $PhCH = O + HCOOH$ |
| (B) 1,2-cyclohexane dione | (Q) $CH_3CH_2CHO + HOOCCH_3$ |
| (C) PhCH (OH) CHO | (R) $HOOC(CH_2)_4COOH$ |
| (D) $CH_3CH_2CH(OH)COCH_3$ | (S) $CH_3COOH + HCOOH$ |

**Q16.** Match the following question :

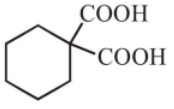
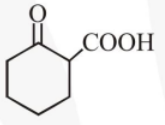
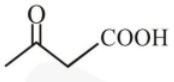
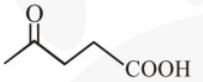
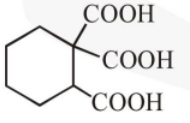
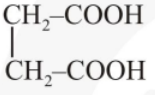
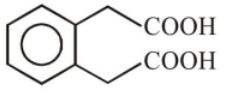
| **Column I (Correct about product)** | **Column II** |
|---|---|
| (A) $R - CN \xrightarrow{H_3O^+}$ | (P) Product is yellow oily liquid |
| (B) $R - NH_2 \xrightarrow[KOH]{CHCl_3}$ | (Q) Gives red colour with CAN |
| (C) $R - NH_2 \xrightarrow[HCl]{NaNO_2}$ | (R) Gives fruity smell with $CH_3OH$ |
| (D) $R_2NH \xrightarrow[HCl]{NaNO_2}$ | (S) Foul smelling compound is formed. |

## NUMERICAL TYPE

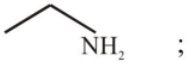
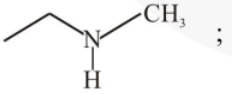
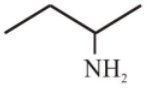
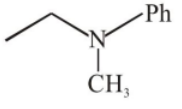
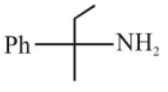
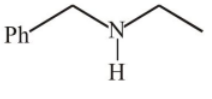
*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

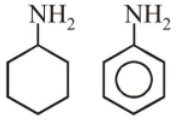
**Q17.**  Number of $CO_2$ evolved.

**Q18.** How many of the following carboxylic acid evolve $CO_2$ on heating :

(a) 
(b) 
(c) 
(d) 
(e) 
(f) 
(g) 

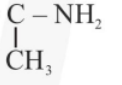
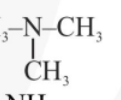
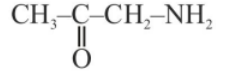
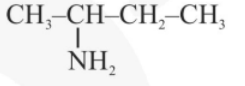
**Q19.** Of the following amines how many can be seperated by Hoffmann's mustard oil reaction.

 ;
 ;
 ;

 ;
 ;


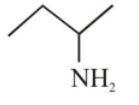
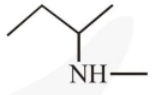
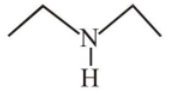
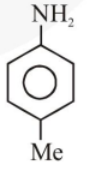
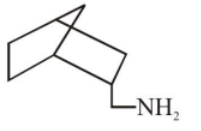
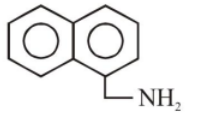
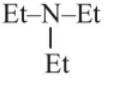
**Q20.** Number of tests by which  can be distinguished.

(i) Isocynide test
(ii) Hinsberg's test
(iii) Musturd oil test
(iv) Dye-azo test
(v) Lucas test
(vi) Victor-mayer's test

**Q21.** How many of the following compounds will give isocyanide (carbylamine) test positive :

(a) $CH_3–CH_2–NH_2$ (b) $CH_2= CH–NH_2$
(c)  (d) $CH_3–CH_2–NH–CH_3$
(e)  (f) 
(g) $Ph–NH_2$ (h) 
(i) 

**Q22.** How many of the following compounds will give isocyanide (carbylamine) test positive :

(a) 
(b) 
(c) 
(d) 
(e) 
(f) 
(g) 
(h) 
(i) 

**Q23.** Which of the following reaction are correctly matches with product :

(a) $CH_3-CH_2-\overset{O}{\overset{\|}{C}}-OH \xrightarrow{PCl_5} H_3C-CH_2-\overset{O}{\overset{\|}{C}}-Cl$

(b) $CH_3-\underset{OH}{\underset{|}{CH}}-COOH \xrightarrow{\Delta} CH_2=CH-COOH$

(c) $H-\overset{O}{\overset{\|}{C}}-OEt \xrightarrow{CH_3MgBr\ (2\ eq.)} CH_3-\underset{OH}{\underset{|}{CH}}-CH_3$

(d) $CH_3-\overset{O}{\overset{\|}{C}}-OEt \xrightarrow{DIBAL\text{-}H} 2CH_3-CH_2-OH$

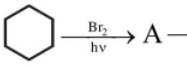
(e) $CH_3-\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}-CH_3 \xrightarrow{PCl_5} 2CH_3-\overset{O}{\overset{\|}{C}}-Cl$

(f) $CH_3-\overset{O}{\overset{\|}{C}}-NH_2 \xrightarrow{NaOH+Br_2} CH_3-CH_2-NH_2$

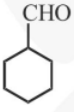
## *Exercise - 2* *Brain Booster*

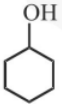
### SINGLE CORRECT TYPE

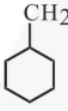
*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

**Q1.**  'C' is :-

(A) 

(B) 

(C) 

(D) 

**Q2.** $\underset{CH_2OH}{\overset{CH_2OH}{|}} \xrightarrow{PBr_3} A \xrightarrow[excess]{KCN} B \xrightarrow{H_3O^+} C$

In the above sequence of reactions the end product (C) is -

(A) $\underset{CH_2CH_2CN}{\overset{CH_2CH_2CN}{|}}$ (B) $\underset{CH_2CN}{\overset{CH_2CN}{|}}$

(C) $\underset{CH_2COOH}{\overset{CH_2COOH}{|}}$ (D) $\underset{CH_2CH_2-NH_2}{\overset{CH_2CH_2-NH_2}{|}}$

**Q3.**  is -

(A) Phenyl ethanoic acid

(B) Benzoic acid

(C) Benzyl amine

(D) Aniline

**Q4.** Consider the following sequence of reactions $CH_2=CH_2 \xrightarrow[CCl_4]{Br_2} A \xrightarrow[2.\ H_3O^+;heat]{1.KCN} B$ the end product (B) is -

(A) Malonic acid

(B) Oxalic acid

(C) Succinic acid

(D) Adipic acid

**Q5.** In the given reaction :

$C_6H_5-\overset{O}{\overset{\|}{C}}-CH_3 \xrightarrow[(ii)\ H^{\oplus}]{(i)\ Br_2/KOH} CHBr_3 + [X]$

[X] will be :

(A) $C_6H_5-CHO$

(B) $C_6H_5COOH$

(C) $C_6H_5-CH_2OH$

(D) $CH_3COOH$

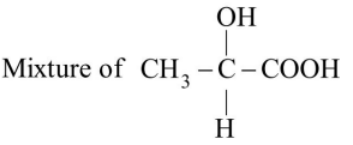
**Q6.** In the reaction sequence:

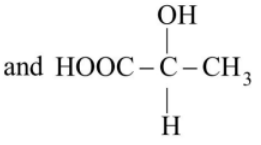
$CH_3-\overset{O}{\overset{\|}{C}}-H \xrightarrow[\overset{\ominus}{O}H]{HCN} (A) \xrightarrow{H_2O/H^{\oplus}}$ Product

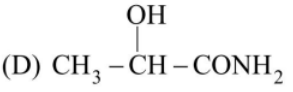
Product will be :

(A) $CH_3-\underset{H}{\overset{OH}{\overset{|}{\underset{|}{C}}}}-COOH$

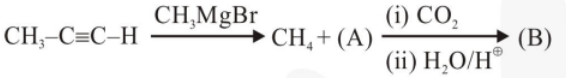
(B) $HOOC-\underset{H}{\overset{OH}{\overset{|}{\underset{|}{C}}}}-CH_3$

(C) Mixture of 

and 

(D) 

**Q7.** In the reaction sequence :



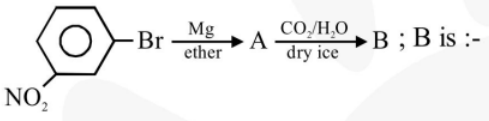
(B) will be :-

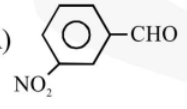
(A) $CH_3–C\equiv C–CH_3$

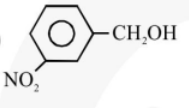
(B) $CH_3–C\equiv C–MgBr$

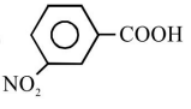
(C) $CH_3–C\equiv C–COOH$

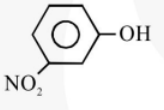
(D) $CH_3–CH=CH–COOH$

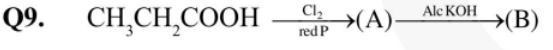
**Q8.**  ; B is :-

(A) 

(B) 

(C) 

(D) 

**Q9.** 

The compound (B) is :-

(A) $CH_3CH_2OH$

(B) $CH_3CH_2COCl$

(C) $CH_2 = CHCOOH$

(D) $CH_3 – CHCl –COOH$

**Q10.** In the given reaction :

[X] + Acetic anhydride ⟶ Aspirin

[X] will be :

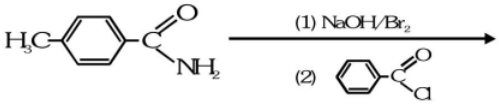
(A) Benzoic acid

(B) *o*-methoxybenzoic acid

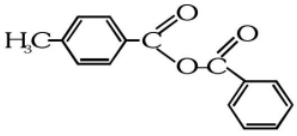
(C) *o*-Hydroxybenzoic acid

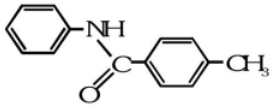
(D) *p*-Hydroxybenzoic acid

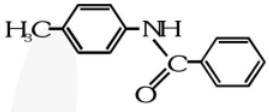
**Q11.** In the reaction

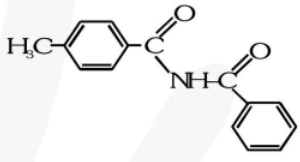


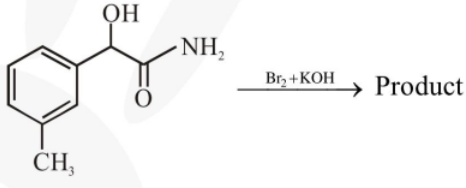
T the structure of the product T is -

(A) 

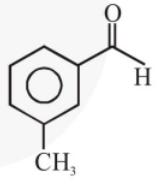
(B) 

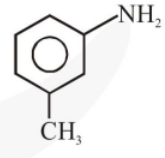
(C) 

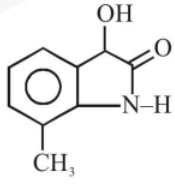
(D) 

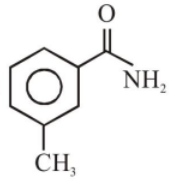
**Q12.** 

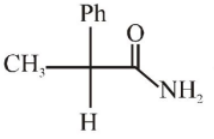
The final product is -

(A) 

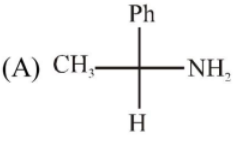
(B) 

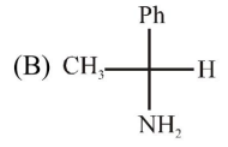
(C) 

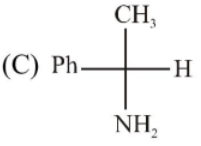
(D) 

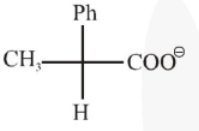
**Q13.**  (A) major

product of the reaction is :-

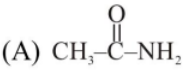
(A) 

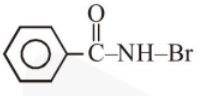
(B) 

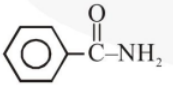
(C) 

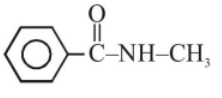
(D) 

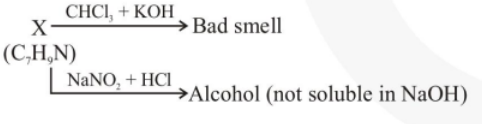
**Q14.** Which of the folloiwng will not give Hoffmann bromamide reaction :-

(A) 

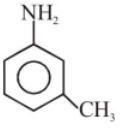
(B) 

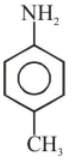
(C) 

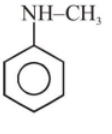
(D) 

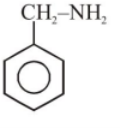
**Q15.** 

X is soluble in HCl. X will be :

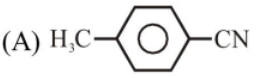
(A) 

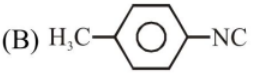
(B) 

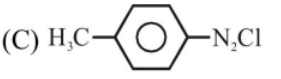
(C) 

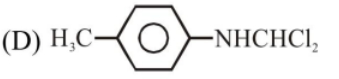
(D) 

**Q16.** The reaction of chloroform with alcoholic KOH and *p*-toluidine forms :

(A) 

(B) 

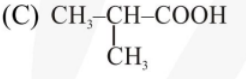
(C) 

(D) 

**Q17.** An organic compound A upon reacting with $NH_3$ gives B. On heating, B gives C. C in presence of KOH reacts with $Br_2$ to give $CH_3CH_2NH_2$. A is :

(A) $CH_3COOH$

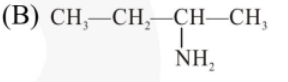
(B) $CH_3CH_2CH_2COOH$

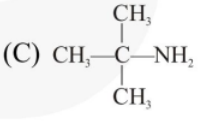
(C) 

(D) $CH_3CH_2COOH$

**Q18.** Which one shows carbyl amine test

(A) $CH_3—CH_2—NH_2$

(B) 

(C) 

(D) All of the above

**Q19.** Benzylamine & aniline can be distinguished by

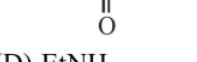
(A) Isocynide test

(B) Mustard oil test

(C) Dye-azo test

(D) Hinsberg test

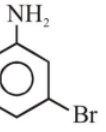
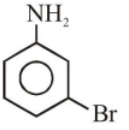
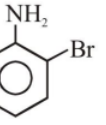
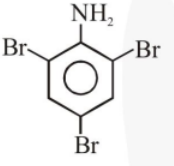
**Q20.** Which of the following can not give carbyl amine reaction -

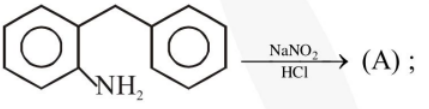
(A) $CH_3NH_2$ (B) 

(C) $PhNH_2$ (D) $EtNH_2$

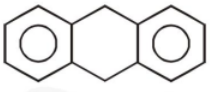
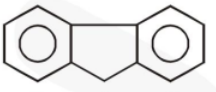
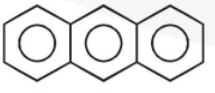
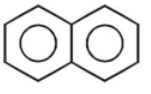
**Q21.** Primary, secondary and tertiary amines can be distinguished by :-
(A) Hinsberg reagent
(B) Tollens reagent
(C) Fehling's solution
(D) a grignard reagent

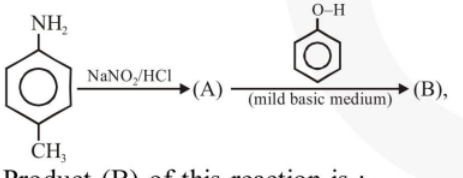
**Q22.** Major product of reaction of aniline with $Br_2/H_2O$.

(A) 
(B) 
(C) 
(D) 

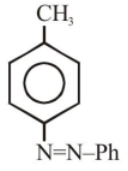
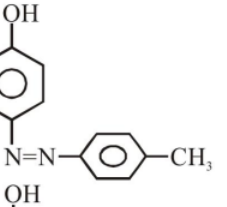
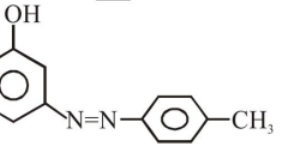
**Q23.** 

Product of this reaction is :

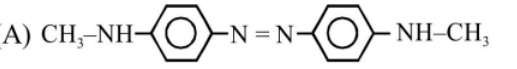
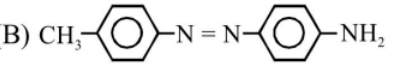
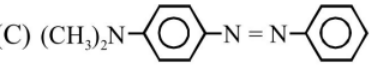
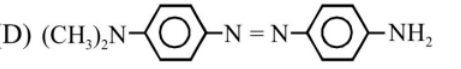
(A) 
(B) 
(C) 
(D) 

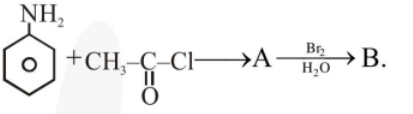
**Q24.** 

Product (B) of this reaction is :-

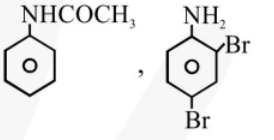
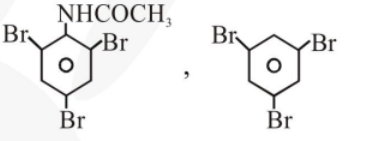
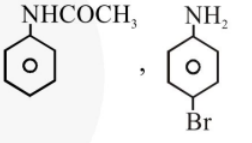
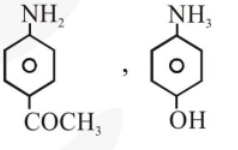
(A) 
(B) 
(C) 
(D) 

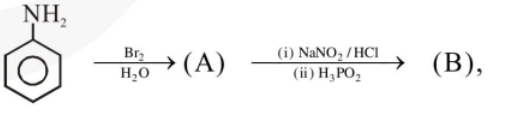
**Q25.** Aniline when diazotised in cold and then treated with dimethyl aniline gives a coloured product. Its structure would be :-

(A) 
(B) 
(C) 
(D) 

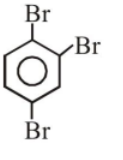
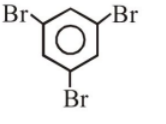
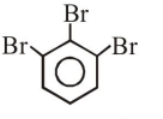
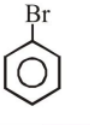
**Q26.** 

A and B are :-

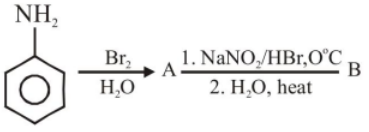
(A) 
(B) 
(C) 
(D) 

**Q27.** 

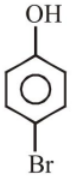
Product (B) in this reaction is :-

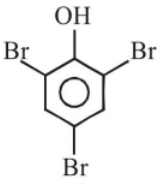
(A) 
(B) 
(C) 
(D) 

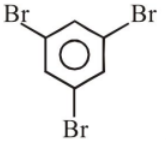
**Q28.** Consider the following sequence of reactions.

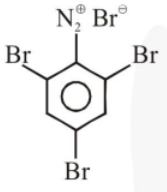


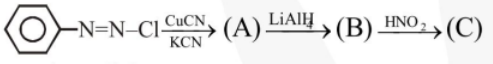
The end product (B) is-

(A) 

(B) 

(C) 

(D) 

**Q29.** 

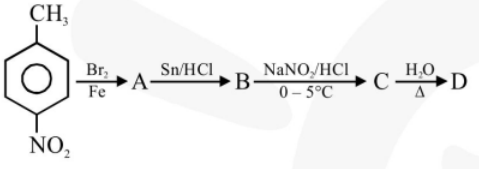
Product C is -

(A) Benzoic acid

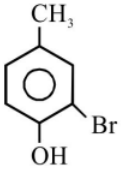
(B) Phenylacetic acid

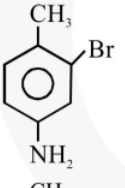
(C) Benzyl alcohol

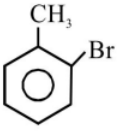
(D) Benzamide

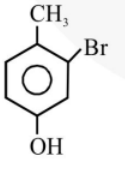
**Q30.** 

what is D :-

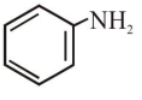
(A) 

(B) 

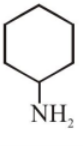
(C) 

(D) 

**Q31.** Which of the following is most basic ?

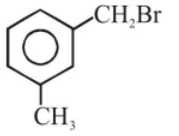
(A) 

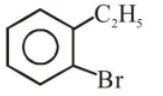
(B) 

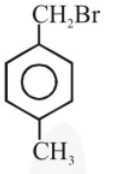
(C) 

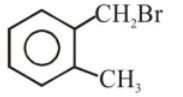
(D) 

**Q32.** Compound (A), $C_8H_9Br$, gives a white precipitate when warmed with alcoholic $AgNO_3$. Oxidation of (A) gives an acid (B), $C_8H_6O_4$. (B) easily forms anhydride on heating. Identify the compound (A):

(A) 

(B) 

(C) 

(D) 

**Q33.** 

X , Y & Z can be :

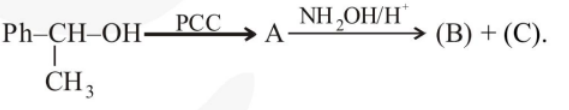
(A) HCl ; $LiAlH_4$ ; $CH_3MgX$

(B) $SOCl_2$ ; $H_2$/Pd $BaSO_4$ ; $(CH_3)_2Cd$

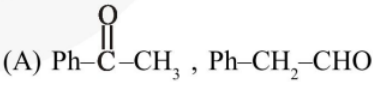
(C) $PCl_5$ ; $NaBH_4$ ; $(CH_3)Li$

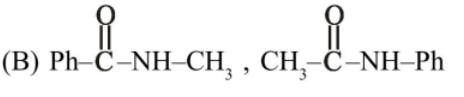
(D) $SOCl_2$ ; $H_2$/Ni ; $(CH_3)_2CuLi$

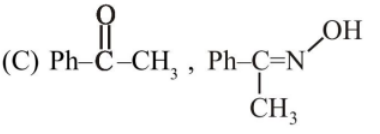
**Q34.** In the given reaction sequence

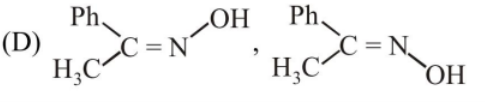


B & C is :-

(A) 

(B) 

(C) 

(D) 

**Q35.** $CH_3-CO_2H$ → (AgOH + $Br_2$) X; (Red P + $Br_2$) Y; ($SOCl_2$ + $CH_2N_2$ + Moist $Ag_2O$ /Δ) Z

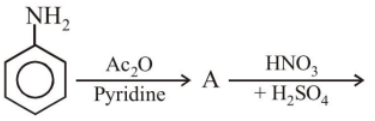
X, Y , Z are respectively :

(A) $CH_3-\overset{\overset{O}{\|}}{C}-O-CH_3$ , $CH_2(Br)-CO_2H$ , $CH_3-\overset{\overset{O}{\|}}{C}-Cl$

(B) $CH_3-Br$ , $CH_3-CH_3$ , $CH_3-CH_2-CO_2H$

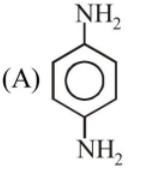
(C) $CH_3-Br$ , $CH_2(Br)-CO_2H$ , $CH_3CH_2CO_2H$

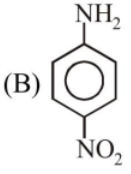
(D) $CH_3COBr$, $CH_3-CH_3$ , $CH_3-\overset{\overset{O}{\|}}{C}-CH_3$

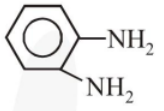
**Q36.** 

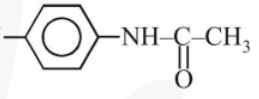
B (Major) + C $\xrightarrow{\text{Chromatographic separation}}$ B $\xrightarrow{H_3O^+}$ D $\xrightarrow[+HCl]{Zn}$ E

E is :

(A) 

(B) 

(C) 

(D) 

## *Exercise - 2A* *Brain Booster*

### ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

**Q1.** Mixture of 1°, 2° and 3° amines can be separated by:

(A) Hinsberg's method

(B) Hofmann's isocyanide test

(C) Fractional distillation

(D) $NaNO_2$ HCl

**Q2.** $\overset{(X)}{C_4H_{11}N} + HNO_2 \longrightarrow C_4H_{10}O$ (3° alcohol)

hence X will give:

(A) Carbyl amine reaction

(B) Hofmann mustard oil reaction

(C) Diazonium salt (as the intermediate) with $HNO_2$

(D) Base insoluble product with Hinsburg reagent

**Q3.** Which of the following amine reacts with Hinsberg reagent to give base soluble product :-

(A) Neopentyl amine

(B) sec propyl amine

(C) diethyl amine

(D) ethyl methyl amine

**Q4.** Which of the following gives silver mirror test?

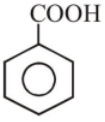
(A) HCOOH

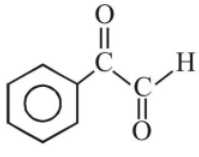
(B) $CH_3COCHOHCH_3$

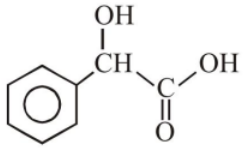
(C) Tartaric acid

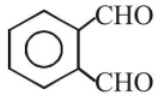
(D) Glucose

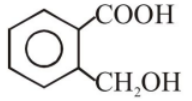
**Q5.** Comp

(A) $C_8H_6O_2 \xrightarrow[(ii)\,H_3O^+]{(i)\,NaOH}$ (B) $\xrightarrow[\Delta]{KMnO_4}$ 

(A) 'A' is 

(B) 'B' is 

(C) 'A' is 

(D) 'B' is 

**Q6.** Which of the following compounds will give acetic acid with $KMnO_4/H^{\oplus}/\Delta$:

(A) $CH_3 - CHO$

(B) $CH_3 - CH = CH - CH_3$

(C) $CH_3 - C \equiv C - CH_3$

(D) $CH_3CH_2OH$

## PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

**Paragraph for Q. 07 to 08**

Two isomeric carboxylic acids H & I ($C_8H_7COOH$) react with $H_2$ / Pd giving compounds $C_8H_9COOH$. H gives a resolvable product whereas I gives a non resolvable product. Both isomers can be oxidised to $PhCO_2H$ by $H^+/KMnO_4$.

**Q7.** Which option is correct regarding I :

(A) I is $PhCH{=}CH{-}CO_2H$

(B) I exist as E, Z diastereomers

(C) I reacts with $NaHCO_3$ to give $CO_2$ gas

(D) All

**Q8.** Which option is correct regarding H :

(A) H is $PhCH{=}CH{-}CO_2H$

(B) H on reduction with $H_2$/Pd gives a product with one chiral centre.

(C) H shows optical isomerism

(D) IUPAC name of H is 2-phenyl prop-3-enoic acid

**Paragraph for Q. 09 to 10**

From the given sequence of reaction,

$$CH_3{-}(CH_2)_4{-}CH_3 \xrightarrow[\Delta]{Al_2O_3,\ Cr_2O_3} P$$

$$CH_3{-}CH_2{-}CH_2{-}\overset{Cl}{\underset{|}{C}}H_2 \xrightarrow[AlCl_3]{} Q \xrightarrow[\Delta]{KMnO_4 / H^+} R \xrightarrow{\text{'X'}} P$$

**Q9.** Reagent 'X' can not be :

(A) $NaHC\overset{18}{O}_3$, $\Delta$

(B) Zn, $\Delta$

(C) NaOH / CaO, $\Delta$

(D) $SeO_2$

**Q10.** Correct statement among the following :

(A) Carbocation is formed as an intermediate from P to Q

(B) From R to P conversion carbanion is formed as an intermediate

(C) Rearrangment is involved during P to Q conversion

(D) Rate for friedal craft alkylation of P product is greater than that of Q product

## MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

**Q11.** Match the reaction given in List I with the information about product given in list II.

| **List-I (Reaction)** | **List-II (Product Information)** |
|---|---|
| (P) $\begin{matrix}CH_2-CH_2-COOH\\ \vert \\ CH_2-CH_2-COOH\end{matrix} \xrightarrow{\Delta}$ | (1) Final product formation involves dehydration only |
| (Q) $\begin{matrix}CH_2-COOH\\ \vert \\ CH_2-COOH\end{matrix} \xrightarrow{\Delta}$ | (2) Final product formation involves decarboxylation only |
| (R) $CH_3-CH_2-\overset{\overset{O}{\Vert}}{C}-CH_2COOH \xrightarrow{\Delta}$ | (3) Final product is an anhydride |
| (S) $CH_3-\overset{\overset{O}{\Vert}}{C}-NH_2 \xrightarrow[\Delta]{P_2O_5}$ | (4) Final product invovles dehydration as well as decarboxylation both |

**Code :**

| | P | Q | R | S |
|---|---|---|---|---|
| (A) | 1 | 4 | 2 | 3 |
| (B) | 4 | 1,3 | 2 | 1 |
| (C) | 2 | 3 | 1 | 4 |
| (D) | 1 | 3 | 4 | 2 |

## MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

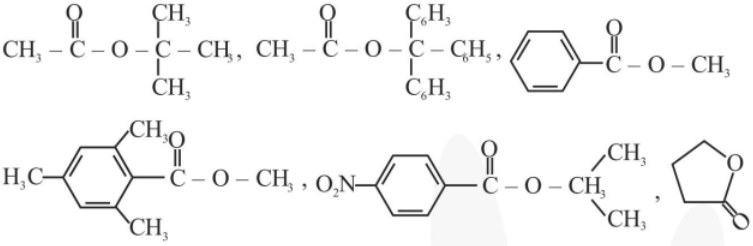
**Q12.** Match the following question :

| **Column I** | **Column II** |
|---|---|
| (A) $RCN \xrightarrow{\text{reduction}}$ | (P) 1° Amine |
| (B) $RCN \xrightarrow[(ii)\,H_2O]{(i)\,CH_3MgBr}$ | (Q) Alcohol |
| (C) $RNC \xrightarrow{\text{hydrolysis}}$ | (R) Ketone |
| (D) $RNH_2 \xrightarrow{HNO_2}$ | (S) Acid |
| | (T) Alkane |

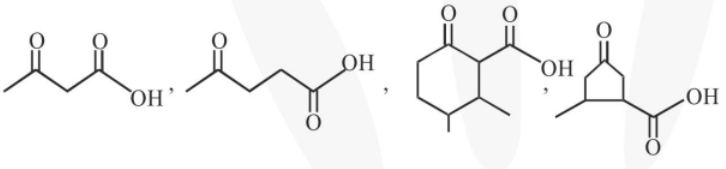
## NUMERICAL TYPE

The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)

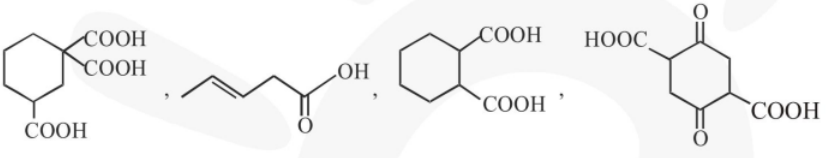
**Q13.** How many of following esters show $A_{AL^{-1}}$ hydrolysis (Acid catayzed, unimolecular and anlkyl-oxygen fission ester hydrolysis)



**Q14.** Examine the structure of following compounds, and find out number of compounds that will undergo decarboxylation in presence of heat.



(a) (b) (c) (d)



(e) (f) (g) (h)

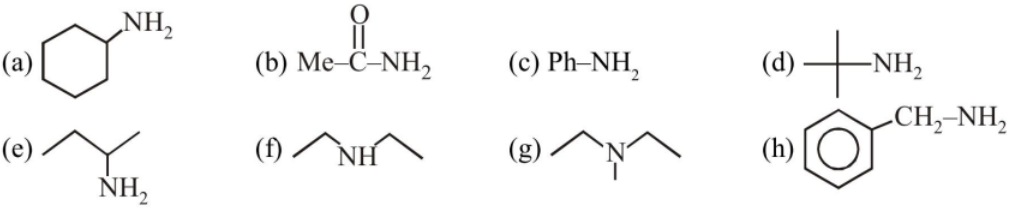
**Q15.** $CH_3CH_2-\underset{\underset{NH_2}{|}}{CH}-COOH \xrightarrow{\Delta} P$ (no. of products)

$CH_3CH_2-\underset{\underset{OH}{|}}{CH}-COOH \xrightarrow{\Delta} Q$ (no. of products)

$CH_3\underset{\underset{OH}{|}}{CH}-CH_2-COOH \xrightarrow{\Delta} R$ (no. of products)

$\underset{\underset{OH}{|}}{CH_2}-CH_2-CH_2-COOH \xrightarrow{\Delta} S$ (no. of products)

**Q16.** How many of the following give positive isocyanide test



(a) (b) $Me-\overset{\overset{O}{\|}}{C}-NH_2$ (c) $Ph-NH_2$ (d)

(e) (f) (g) (h)

# Answer Key

| Ex-1 | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| B | A | A | B | B | B | C | D | D | B |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| D | B | A | B | C | B | B | B | B | D |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | |
| C | B | D | B | B | D | A | C | A | |

## Ex-1A

**1.** A,C **2.** B,C **3.** A,C,D **4.** A,B,C 5. A,C,D
**6.** C,D **7.** B,D **8.** A,B,C,D **9.** D **10.** A
**11.** B **12.** B 13. D
**14.** A→P, S ; B→Q, S ; C→P, S ; D→R **15.** A→S ; B→R ; C→P ; D→Q
**16.** A→R ; B→S ; C→Q ; D→P **17.** 4 **18.** 5
**19.** 4 **20.** 1 **21.** 6 **22.** 5 **23.** 3

| Ex-2 | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| A | C | A | C | B | C | C | C | C | C |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| C | A | A | D | D | B | D | D | C | B |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| A | D | B | B | C | C | B | B | D | D |
| 31 | 32 | 33 | 34 | 35 | 36 | | | | |
| C | D | B | B | C | A | | | | |

## Ex-2A

**1.** A, C **2.** A, B, C **3.** A, B **4.** A, B, D **5.** A, B
**6.** A, B, C **7.** D **8.** B **9.** A,B,D **10.** A, B, C
**11.** B **12.** A→P ; B→R ; C→P, S ; D→Q
**13.** 3 **14.** 5 **15.** 9 **16.** 5